

Relations Solutions

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Exercises

1. Define two points (x_0, y_0) and (x_1, y_1) of the plane to be equivalent if $y_0 - x_0^2 = y_1 - x_1^2$. Check that this is an equivalence relation and describe the equivalence classes.

Proof. For the relation described in this problem to be an equivalence relation, it must be reflexive, symmetric, and transitive. Let $(x, y) \in \mathbb{R}^2$, and note $y - x^2 = y - x^2$ since x and y are the same. Therefore, the relation is reflexive. Let $(x_0, y_0), (x_1, y_1) \in \mathbb{R}^2$ such that $(x_0, y_0) = (x_1, y_1)$, thus $y_0 - x_0^2 = y_1 - x_1^2$. Note that "=" as used in this equality for real numbers is symmetric and therefore $y_1 - x_1^2 = y_0 - x_0^2$. Hence, $(x_1, y_1) = (x_0, y_0)$ and we can conclude our relation is symmetric. Let $(x_0, y_0), (x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ such that $y_0 - x_0^2 = y_1 - x_1^2$ and $y_1 - x_1^2 = y_2 - x_2^2$. Since "=" as used here is the usual equality for real numbers, we note that it is transitive. As such, $y_0 - x_0^2 = y_2 - x_2^2$ and we conclude $(x_0, y_0) = (x_2, y_2)$. Therefore, our relation is transitive and we can conclude that the relation described in this problem is an equivalence relation.

We now examine what the equivalence classes are: They are parabolas. The points on a parabola will all be equivalent to each other under this equivalence relation. In particular, the parabola must be of the form $f(x) = x^2 + c$ for some constant $c \in \mathbb{R}$. Note these classes cover all points on the plane and are disjoint of each other. Further, that points on parabolas of forms other than this one, are already in some parabola $f(x) = x^2 + c$ and that also have a difference that is dependent on x in some way, i.e. $y - x^2 = a^2 + bx$ in such case each point on the parabola will have a difference that is not equal to another point in the parabola except possibly its reflection across the y -axis if $b = 0$ and $a \neq 1$, but in such case this pair of points is already in another equivalence class, mainly $f(x) = x^2 + c$. $\square$

2. Let C be a relation on a set A . If $A_0 \subseteq A$, define the **restriction** of C to A_0 to be $C \cap (A_0 \times A_0)$. Show that the restriction of an equivalence relation is an equivalence relation.

Proof. Let $C_0 = C \cap (A_0 \times A_0)$. If $A_0 = \emptyset$, then $C_0 = \emptyset$ also. Note if A is a non-empty set, then C_0 is not an equivalence relation on A since it is not reflexive. If, however, A is empty, then C_0 is vacuously an equivalence relation on A . Thus, we will assume that A_0 is non-empty. We begin by showing that C_0 must be reflexive. Let $x \in A_0$, then $(x, x) \in A_0 \times A_0$. Since C is an equivalence relation on A we have that $(x, x) \in C$ by definition. Thus, $(x, x) \in C_0$ and we can conclude that C_0 is reflexive on A_0 . To show that C_0 is symmetric, let $(x, y) \in C_0$. Thus, $(x, y) \in C$ and $(x, y) \in A_0 \times A_0$. This implies that $x, y \in A_0$ and thus $(y, x) \in A_0 \times A_0$ also. Furthermore, since C is a relation on A , $(y, x) \in C$. Thus, $(y, x) \in C_0$ if $(x, y) \in C_0$, showing that C_0 is symmetric. Finally, to

show that C_0 is transitive, let $(x, y), (y, z) \in C_0$. Note that $x, y, z \in A_0$ by construction. Hence, $(x, z) \in A_0 \times A_0$. Since C is a relation on A , it must be that $(x, z) \in C$. Therefore, $(x, z) \in C_0$, showing that C_0 is transitive and thus is an equivalence relation on A_0 . $\square$

3. Here is a "proof" that every relation C that is both symmetric and transitive is also reflexive: "Since C is symmetric, aCb implies bCa . Since C is transitive, aCb and bCa together imply aCa , as desired." Find the flaw in this argument.

The main flaw with this argument is that reflexivity requires all elements in a set to be included in the relation, i.e., $(x, x) \in R$ for all $x \in A$, while symmetry and transitivity do not have this requirement. In particular, say you have the set $A = \{a, b, c, d\}$ with the relation $R = \{(a, b), (b, a), (b, c), (c, b), (a, c), (c, a), (a, a), (b, b), (c, c)\}$. Note that R is symmetric and transitive; however, it is not reflexive since $(d, d) \notin R$.

4. Let $f : A \rightarrow B$ be a surjective function. Let us define a relation on A by setting $a_0 \sim a_1$ if

$$f(a_0) = f(a_1).$$

(a) Show that this is an equivalence relation.

Proof. Note that for $a \in A$, $f(a) = f(a)$, thus $a \sim a$ for all $a \in A$. Let $x, y \in A$ such that $f(x) = f(y)$. Since $=$ is assumed to be an equivalence relation on B , it must be that $f(y) = f(x)$, thus showing that $y \sim x$ if $x \sim y$. Let $x, y, z \in A$ such that $x \sim y$ and $y \sim z$. By definition, we have that $f(x) = f(y)$ and $f(y) = f(z)$. Since $=$ is an equivalence relation on B we have that $f(x) = f(z)$. Therefore, $x \sim z$, and we conclude that $\sim$ is an equivalence relation on A . $\square$

(b) Let A^* be the set of equivalence classes. Show that there is a bijective correspondence of A^* with B .

Proof. Let $g : A^* \rightarrow B$ be a function such that $g(a^*) = b$ where a^* is an equivalence class in A determined by a , and $b = f(a)$. We claim that g is bijective. Consider $b \in B$, since f is surjective, there exists some $a \in A$ such that $f(a) = b$. Hence, $f(a^*) = b$ by construction and we can conclude that g is surjective. Let $g(x^*) = b$ and $g(y^*) = b$. By construction, $f(x) = b$ and $f(y) = b$. Note this implies that $x \sim y$ and thus both are elements of the same equivalence class, i.e., $x^* = y^*$ since equivalence classes are either disjoint or equal. Therefore, g is injective, and we can conclude that g is bijective. $\square$

5. Let S and S' be the following subsets of the plane:

$$S = \{(x, y) \mid y = x + 1 \text{ and } 0 < x < 2\},$$

$$S' = \{(x, y) \mid y - x \text{ is an integer}\}.$$

- (a) Show that S' is an equivalence relation on the real line and $S' \supseteq S$. Describe the equivalence classes of S' .

Proof. Let $x \in \mathbb{R}$ and note that $x - x = 0$ which is an integer. Thus, $xS'x$ for all $x \in \mathbb{R}$. Let $xS'y$, this implies that $y - x = n$ for some integer n . Note we can rewrite this equation as $x - y = -n$ where $-n$ is still an integer, and thus, $yS'x$. Let $xS'y$ and $yS'z$, then $y - x = n$ and $z - y = m$ for integers n and m . Note these equations imply $z - x = m + n$. Since $n + m$ is an integer, we conclude that $xS'z$. Therefore, S' is an equivalence relation on the real line.

Let $(x, y) \in S$, then $y - x = 1$ where $0 < x < 2$. Since $y - x$ is equal to an integer, they belong to an equivalence class in S' . Thus, $(x, y) \in S'$, and we can therefore conclude that $S \subseteq S'$.

In regard to the equivalence classes of S' , note that all the integers are in the same equivalence class as 0, since their difference with zero is always an integer. Every rational number between zero and one are in their own distinct equivalence class, since rational numbers between zero and one are all less than one away from each other. This is also true for the irrational numbers that are between zero and one. In particular, the numbers between zero and one, including zero also, can determine the equivalence classes of this equivalence relation. As such, we could represent the equivalence classes of this equivalence relation as $[0, 1)$. $\square$

- (b) Show that given any collection of equivalence relations on a set A , their intersection is an equivalence relation on A .

Proof. Let $R = \bigcap_{i \in I} R_i$ where R_i are equivalence relations on A and I is some indexing set. Let $x \in A$, since every R_i is an equivalence relation it must be that $(x, x) \in R_i$ for all $i \in I$. Thus, $(x, x) \in R$ by definition. Let $(x, y) \in R$, this implies that $(x, y) \in R_i$ for all i . Since each R_i is an equivalence relation, $(y, x) \in R_i$ for all i . Hence, $(y, x) \in R$. Let $(x, y), (y, z) \in R$. This implies that $(x, y), (y, z) \in R_i$ for all i . Since each R_i is an equivalence relation, it must be that $(x, z) \in R_i$ for all i . Therefore, $(x, z) \in R$ and we can thus conclude that R is an equivalence relation on A . $\square$

- (c) Describe the equivalence relation T on the real line that is the intersection of all equivalence relations on the real line that contain S . Describe the equivalence classes of T .

Proof. One way to find T is to start with the set S , and construct an equivalence class around S . As a first step, we will have that $S \subseteq T$. This is not an equivalence class since S is not reflexive, transitive, nor symmetric. Thus, we must include the set $S_R = \{(x, x) \mid x \in \mathbb{R}\}$. Note $S \cup S_R \subseteq T$ is now reflexive. To make this relation symmetric, we can look back at the definition of S and construct a set that is symmetric to S . Recall $S = \{(x, y) \mid y = x + 1 \text{ and } 0 < x < 2\}$. A set that is symmetric to S can be defined as $S_S = \{(x, y) \mid y = x - 1 \text{ and } 1 < x < 3\}$. Thus, the set $S \cup S_R \cup S_S \subseteq T$ is now reflexive and symmetric. We need only make this set transitive to truly make it an equivalence relation. Note there are elements in S that should provide other points to be included through transitivity, like $(0.5, 1.5)$ and $(1.5, 2.5)$. These points should imply that $(0.5, 1.5)$ to also be included. Note that this can be accomplished by including the set $S_{T_1} = \{(x, y) \mid y = x + 2 \text{ and } 0 < x < 1\}$. Since we included this set, we must include a set symmetric to it, $S_{T_2} = \{(x, y) \mid y = x - 2 \text{ and } 2 < x < 3\}$. Thus, note that $S \cup S_R \cup S_S \cup S_{T_1} \cup S_{T_2} = T$ is a reflexive, symmetric, and transitive subset of $\mathbb{R} \times \mathbb{R}$, and is therefore an equivalence relation. Note T must be the intersection of all equivalence relations that include S since it is the smallest equivalence relation that includes S by construction.

In regard to the equivalence classes of T , note that if $x < 0$ or $x > 3$, its equivalence class is the set $\{x\}$ since it is only equivalent to itself by construction. If $0 < x < 1$, then its equivalence class is the set $\{x, x + 1, x + 2\}$ by construction. If $1 < x < 2$, then its equivalence class is the set $\{x - 1, x, x + 1\}$ by construction. Finally, if $2 < x < 3$, then its equivalence class is the set $\{x - 2, x - 1, x\}$ by construction. $\square$

6. Define a relation on the plane by setting

$$(x_0, y_0) < (x_1, y_1)$$

if either $y_0 - x_0^2 < y_1 - x_1^2$, or $y_0 - x_0^2 = y_1 - x_1^2$ and $x_0 < x_1$. Show that this is an order relation, and describe it geometrically.

Proof. Let $(x, y), (n, m)$ be distinct points on the plane. This implies that either $x \neq n$, $y \neq m$, or both. If $x \neq n$, either $x^2 < n^2$ or $x^2 > n^2$ or $x^2 = n^2$. Without loss of generality let $x^2 < n^2$. Since $y = m$, we note that $y - x^2 > m - n^2$. Hence $(x, y) > (n, m)$. If $y \neq m$, then either $y < m$ or $y > m$. Without loss of generality let $y < m$. Since $x = n$, this implies that $y - x^2 < m - n^2$. Hence, $(x, y) < (n, m)$. If both $x = n$ and $y = m$ it may be that $y - x^2 = m - n^2$. Note then that $x \neq n$ and thus either $x < n$ or $x > n$, in either case we can still compare both points of the plane. Therefore, the relation in this exercise is comparable.

Let (x, y) be a point on the plane. Note that $y - x^2 = y - x^2$ and that $x = x$. Hence, our relation for this exercise does not hold for this point and itself. In particular, it implies that a point is not comparable with itself. Therefore, our relation is non-reflexive.

Let $(x, y) < (u, v)$ and $(u, v) < (s, t)$. Then, by definition, $y - x^2 < v - u^2$ or $y - x^2 = v - u^2$ and $x < u$. Similarly, $v - u^2 < t - s^2$ or $v - u^2 = t - s^2$ and $u < s$. Without loss of generality

let $y - x^2 < v - u^2$ and $v - u^2 < t - s^2$. This implies that $y - x^2 < t - s^2$, showing that $(x, y) < (s, t)$ as desired. Therefore, this relation is transitive and we can conclude that this is an order relation.

Geometrically, we note that every point on the plane will lie in exactly one parabola of the form $y = x^2 + c$ for some $c \in \mathbb{R}$. In particular, by our choice of relation we have made it so that if one point lies in a parabola higher up than another, then that point will be greater than the other. If two points lie on the same parabola, then the point most to the right will be the greater of the two. $\square$

7. Show that the restriction of an order relation is an order relation.

Proof. Let C be an order relation on the set A , and $A_0 \subseteq A$ be a subset of A . Consider the relation $C_0 = C \cap (A_0 \times A_0)$. If $A_0 = \emptyset$ note that $C_0 = \emptyset$. It is vacuously true that C_0 is an order relation on A_0 . Thus, we will assume that A_0 is non-empty. Let $x, y \in A_0$ such that $x \neq y$. Since $A_0 \subseteq A$ it must be that $x, y \in A$. Therefore, either xCy or yCx since C is an order relation. Note that $(x, y), (y, x) \in A_0 \times A_0$ and thus C will contain either (x, y) or (y, x) depending on whichever element C contains. In either case, we can conclude that C_0 is comparable in A_0 . Now, let $x \in A_0$, which implies that $x \in A$. Since C is an order relation on A , this implies that $(x, x) \notin C$. Hence, $(x, x) \notin C_0$ by construction. Therefore, C_0 must be non-reflexive. Let $(x, y), (s, t) \in C_0$. This implies that $(x, y), (s, t) \in C$. Since C is an order relation on A , $(x, t) \in C$ must be true. Since $x, t \in A_0$, $(x, t) \in A_0 \times A_0$ and therefore $(x, t) \in C_0$. Thus we conclude that C_0 is transitive and therefore an order relation. $\square$

8. Check that the relation defined in Example 7 is an order relation.

Proof. The order relation in Example 7 is the following: For the set of real numbers, define xCy if $x^2 < y^2$, or if $x^2 = y^2$ and $x < y$.

We begin by showing that this relation is comparable. Let $x, y \in \mathbb{R}$ such that $x \neq y$. Since Note either $x^2 < y^2$, $x^2 > y^2$ or $x^2 = y^2$. In the first two cases we can conclude that either xCy or yCx showing our relation is comparable. In the third case, we then check if $x < y$ or $x > y$ since $x \neq y$. Note that in either case we can conclude that xCy or yCx showing or relation is comparable. Hence, we conclude that this relation is comparable on the real numbers.

Now we demonstrate that the relation is non-reflexive. Let $x \in \mathbb{R}$, then $x^2 = x^2$ and $x = x$. Note that based on our definition for our relation we cannot relate x to itself. Thus, our relation does not compare elements to themselves and we can conclude that it is non-reflexive.

To show that the relation is transitive, let xCy and yCz . By definition, this implies that either $x^2 < y^2$, or $x^2 = y^2$ and $x < y$. Similarly, we have that $y^2 < z^2$, or $y^2 = z^2$ and $y < z$. If $x^2 < y^2$ and $y^2 < z^2$ note that $x^2 < z^2$ must be true and we conclude that xCz . If $x^2 = y^2$ and $y^2 < z^2$, note that $x^2 < z^2$ must be true and we conclude that xCz . If $x^2 = y^2$ and $y^2 = z^2$, then note $x < y$ and $y < z$. This implies that $x < z$, so we conclude that

$x Cz$. The remaining case is analogous to the one we did earlier but the signs are swapped. Hence, we conclude that our relation is transitive. Therefore, C is an order relation on the real numbers. $\square$

9. Check that the dictionary order is an order relation.

Proof. Let A and B be sets such that $<_A$ and $<_B$ are order relations in their respective sets. We define the dictionary order on $A \times B$ as the relation where $(a, b) < (c, d)$ if $a <_A c$, or if $a = c$ and $b <_B d$. To show that the dictionary order is an order relation we must show that it has comparability, non-reflexivity, and transitivity.

We begin by showing that this relation is comparable. Let $(a, b), (c, d) \in A \times B$ such that $(a, b) \neq (c, d)$. If $a <_A c$, then $(a, b) < (c, d)$, and if $c <_A a$ then $(c, d) < (a, b)$. If $a = c$, it must be that $b \neq d$ by construction. Thus, either $b <_B d$ or $d <_B b$. In either case we can compare both elements of $A \times B$. Therefore, we conclude that the dictionary order is an order relation.

We now demonstrate that the dictionary order is non-reflexive. Let $(x, y) \in A \times B$. Then, $x = x$ and $y = y$. Note that since $y = y$, we cannot compare (x, y) to itself by the definition of the dictionary order. Hence, an element in $A \times B$ is not comparable to itself under this relation. Therefore, we conclude that the dictionary order is non-reflexive.

Finally, to show that the dictionary order is transitive let $(a, b) < (c, d)$ and $(c, d) < (n, m)$. By definition, either $a <_A c$ or $a = c$ and $b <_B d$. Similarly, either $c <_A n$ or $c = n$ and $d <_B m$. If $a <_A c$ and $c <_A n$, then $a <_A n$ by the fact $<_A$ is an order relation in A . Thus we conclude that $(a, b) < (n, m)$. If $a = c$ and $c = n$, it must be that $b <_B d$ and $d <_B m$. Since $<_B$ is an order relation in B it must be that $b <_B m$. Thus, $(a, b) < (n, m)$. If $a = c$ and $d <_B m$ it must be that $b <_B d$. This implies that $(a, b) < (n, m)$. If $a <_A c$ and $c = n$, we conclude that $a <_A n$, showing that $(a, b) < (n, m)$. Therefore, the dictionary order is transitive and we further conclude that it is an order relation on $A \times B$. $\square$

10.

(a) Show that the map $f : (-1, 1) \rightarrow \mathbb{R}$ of Example 9 is order preserving.

Proof. The function in Example 9 is the following: $f : (-1, 1) \rightarrow \mathbb{R}$ such that

$$f(x) = \frac{x}{1 - x^2}.$$

To show that f is order preserving, we must demonstrate that for $x, y \in (-1, 1)$ such that $x < y$, it must be that $f(x) < f(y)$. Thus, assume that $x, y \in (-1, 1)$ such that $x < y$. Consider

$$\frac{y}{1 - y^2} - \frac{x}{1 - x^2}.$$

We need only show that this value is greater than zero. Note that it is equivalent to

$$\frac{y(1 - x^2) - x(1 - y^2)}{(1 - y^2)(1 - x^2)}.$$

The denominator of this fraction is always positive due to the values x and y are allowed to take. As such, we must show that the numerator must also be positive given $x < y$. $\square$

- (b) Show that the equation $g(y) = 2y/[1 + (1 + 4y^2)^{1/2}]$ defines a function $g : \mathbb{R} \rightarrow (-1, 1)$ that is both a left and a right inverse for f .

11. Show that an element in an ordered set has at most one immediate successor and at most one immediate predecessor. Show that a subset of an ordered set has at most one smallest element and at most one largest element.

12. Let $\mathbb{Z}_+$ denote the set of positive integers. Consider the following order relations on $\mathbb{Z}_+ \times \mathbb{Z}_+$:

- (i) The dictionary order.
- (ii) $(x_0, y_0) < (x_1, y_1)$ if either $x_0 - y_0 < x_1 - y_1$, or $x_0 - y_0 = x_1 - y_1$ and $y_0 < y_1$.
- (iii) $(x_0, y_0) < (x_1, y_1)$ if either $x_0 + y_0 < x_1 - y_1$, or $x_0 + y_0 = x_1 + y_1$ and $y_0 < y_1$.

In these order relations, which elements have immediate predecessors? Does the set have a smallest element? Show that all three order types are different.

13. Prove the following:

Theorem. *If an ordered set A has the least upper bound property, then it has the greatest lower bound property.*

14. If C is a relation on a set A , define a new relation D on A by letting $(b, a) \in D$ if $(a, b) \in C$.

- (a) Show that C is symmetric if and only if $C = D$.
- (b) Show that if C is an order relation, D is also an order relation.
- (c) Prove the converse of the theorem in Exercise 13.

15. Assume that the real line has the least upper bound property.

- (a) Show that the sets

$$[0, 1] = \{x \mid 0 \leq x \leq 1\},$$

$$[0, 1) = \{x \mid 0 \leq x < 1\}$$

have the least upper bound property.

- (b) Does $[0, 1] \times [0, 1]$ in the dictionary order have the least upper bound property? What about $[0, 1] \times [0, 1)$? What about $[0, 1) \times [0, 1]$?